

Mandarin Third Tone Sandhi in ‘Unstructured’ Strings: A Stochastic Match Theory Analysis

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Abstract

This article gives an analysis of Mandarin Chinese third tone sandhi. We focus on the foundations of third tone sandhi by concentrating on strings of digits and lists, commonly referred to as ‘unstructured’ strings, whose sandhi patterns are not influenced by morphosyntactic structure. We introduce data from a series of elicitation experiments that show, contrary to popular belief, that such strings in fact manifest internal structure and vary radically in terms of the sandhi patterns they exhibit. We analyze a variety of these patterns using a combination of Match Theory and Stochastic Optimality Theory. The former enables the description of the interplay between sandhi domains, metrical feet, and the aforementioned structural properties of the strings, while the latter is a probabilistic model well-suited for modeling interactions among competing factors of different strengths and describing the asymmetric patterns of variation like those found in our empirical data.

Keywords: Mandarin Chinese, third tone sandhi, prosodic grouping, Optimality Theory, stochastic grammar, Match Theory

1 Introduction

This paper proposes an analysis of Mandarin third tone sandhi based on the Stochastic Optimality Theory of Boersma (1997, 1998) and Boersma and Hayes (2001) and the Match Theory of Selkirk (2011). We focus our attention on expressions like strings of digits and lists to show that, contrary to popular belief, these expressions are by no means unstructured and indeed exhibit radically different behavior with respect to the sandhi patterns they manifest depending on their internal form.

Section 2.1 gives an overview of Mandarin third tone sandhi. In section 2.2, previous analyses of third tone sandhi are discussed, especially that of M. Chen (2000), whose analysis is couched in an Optimality Theory framework. Section 2.3 discusses challenges to the previously discussed analyses posed by experimental data, most pertinently that they

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all fail to recognize that the expressions under discussion do indeed have some internal structure, which is determined by the presence or absence of identical, adjacent elements and the position of those elements (e.g., AAB versus ABB or ABC). In section 3 those experiments and resulting data are described in detail. Section 4 introduces Stochastic Optimality Theory and shows how constraints based on Match Theory can describe the interaction between metrical feet, sandhi domains, and constituents and can provide coverage for the data that is both more empirically accurate and more theoretically sound.

2 Background

2.1 Mandarin tones and third tone sandhi

Most syllables of Mandarin bear one of four lexical tone contours. The typical romanization (*pinyin*) transcription employs four distinct diacritics to represent each one.

Tone	Name	Pinyin	Example
1st	high	mā	媽 (mother)
2nd	rising	má	麻 (hemp)
3rd	dipping	mǎ	馬 (horse)
4th	falling	mà	罵 (scold)

Figure 1: Mandarin lexical tones

Mandarin’s third tone is subject to phonological alternations and can have several different surface forms. The most common is what Cheng (1973, 43-44), citing Chao (1948) calls the ‘half third tone rule,’ whereby the duration and contour of a third-tone syllable are abbreviated (as approximately shown in Figure 2) when it occurs before “any tone other than third tone” but not “at the end of a sentence or before a pause.”

The subject of this paper is a second alternation, commonly known as *third tone sandhi* (henceforth 3TS). This alternation occurs when there is a sequence of two or more consecutive third tones and “all the third tones except for the last one are simultaneously changed into second tone” (*Ibid.* p. 48). As a result of 3TS, a phrase consisting of two syllables with underlying third tones like 好 (*hǎo* ‘good’) and 酒 (*jǐu* ‘wine’) will surface with a second tone on the first syllable: 好酒 (*háo jǐu* ‘good wine’). To complicate matters, an additional phonological alternation is observed in fast speech, whereby “a second tone becomes first when preceded by first or second tone and followed by any tone other than the neutral tone” (*Ibid.*, p. 44). Note that while this process would not operate directly on third tones, it could certainly affect their pronunciation, since 3TS outputs second tones. The two alternations just described could be captured with the ordered rules in (1), with (i) applying before (ii).

- (1) i. 3RD TONE SANDHI RULE: $3 \rightarrow 2 / ___3$
- ii. 2ND TONE SANDHI RULE: $2 \rightarrow 1 / \{1, 2\} ___{\{1, 2, 3, 4\}}$ (in fast speech)

If the rules in (1) were an adequate and complete description of Mandarin tone sandhi, the issue would have been settled long ago and received little attention in recent years. This

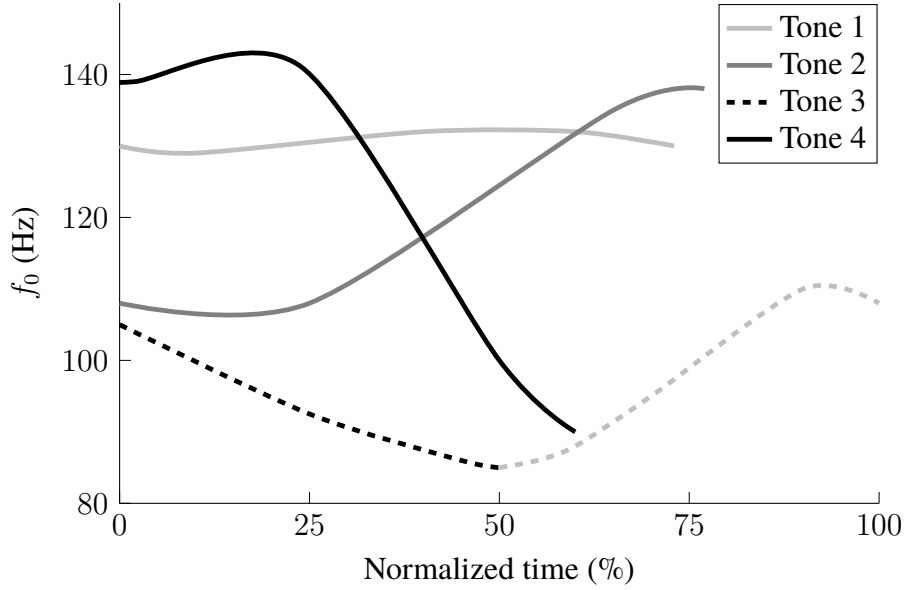


Figure 2: Contours of the four Mandarin tones, based on Lai and Li (2022), adapted from Xu (1997); the grayed-out portion of Tone 3 illustrates the typical truncation of that tone in most phonological contexts.

is far from true. Debates continue to this day about the true nature of the underlying form of the third tone, the actual phonetic character of the output(s) of 3TS, and the domain of application of 3TS and the conditions under which it is triggered. The last issue is the most complex and the only one we will be concerned with in this article. To abstract away from the different phonetic alternants that can arise when one third tone occurs before another, we will use the common notation ‘s’ to indicate any sandhi tone. The most central question is thus:

(2) The Mandarin 3TS Question: $3 \rightarrow s / ?$

Where one or more sandhi tones appear consecutively to the left of a triggering third tone, the sequence $(s_1 \dots s_n 3)$ is said to be a *sandhi domain*. The famous example below illustrates the variable application of 3TS, as well as its sensitivity to morphosyntactic factors. In (3), 好 (*hǎo*) is an adjective modifying the object, while in (4) it is a resultative complement forming a constituent with the verb.

- | | |
|---|---|
| <p>(3) 老 李 買 好 酒 了。</p> <p>lǎo lǐ mǎi hǎo jiǔ le</p> <p>old Lee buy good wine cs</p> <p>‘Old Lee buys good wine now.’</p> | <p>(4) 老 李 買 好 酒 了。</p> <p>lǎo lǐ mǎi hǎo jiǔ le</p> <p>old Lee buy RES wine cs</p> <p>‘Old Lee has now bought wine.’</p> |
|---|---|

As shown in Table 1, the first two tone sandhi patterns, $(s3)3(s3)$ and $(ss3)(s3)$ are available to (3) but unavailable to (4). The discrepancy is due to the fact that the final sandhi domain in both patterns coincides with the phrase boundaries of the object in (3), but does not do so in (4), where it instead would cut between the verb and its resultative complement, creating an awkward domain consisting of two syllables that are not a constituent.

<i>lǎo lǐ</i> [<i>mǎi</i> [<i>hǎo jiǔ</i>]]	tone sequence	<i>lǎo lǐ</i> [<i>mǎi hǎo</i>][<i>jiǔ</i>]
possible	(s 3) 3 (s 3)	not possible
	(s s 3) (s 3)	
	(s 3) (s s 3)	possible
	(s s s s 3)	

Table 1: Tone sandhi in the famous ‘Old Lee’ example

Examples like this have inspired a great amount of research investigating the nuances of 3TS and restrictions on sandhi domains.

2.2 Previous descriptions and analyses of the 3TS domain

2.2.1 3TS in hierarchical structures

There is a well-known discrepancy between 3TS patterns in left-branching versus right-branching constructions.

- (5) 打 狗 好。
dǎ gǒu hǎo
beat dog good
‘Beating dogs is good.’

- (6) 馬 好 渴。
mǎ hǎo kě
horse good thirsty
‘The horse is really thirsty.’

Left-branching constructions like (5) almost always surface as (ss3). Right-branching constructions like (6) almost always surface as 3(s3).¹ The typical explanation for this is that sandhi proceeds bottom-up, node-by-node, and, in a right-branching construction like (6), after the first node has undergone sandhi, no further sandhi is triggered since 3+(s3) would leave no remaining adjacent third tones. In contrast, this is exactly what would happen in (5): the adjacent third tones in (s3)+3 would trigger a second application of 3TS, resulting in a trisyllabic sandhi domain. Cf. (a) and (b) in Figure 3.

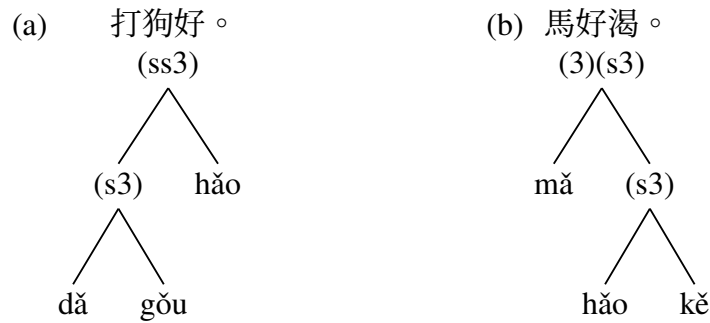


Figure 3: 3TS in left- and right-branching structures, examples (5) and (6) respectively

¹As an illustration, in our own elicitation experiment, (5) was pronounced as (ss3) by 33 out of 35 subjects and (6) was pronounced as (ss3) by zero of them.

2.2.2 3TS in unstructured strings

In strings without any internal hierarchical organization, 3TS is typically assumed (e.g., C. Shih (1986, p. 110 and 1997, p. 91) or M. Chen (2000, p. 366)) to apply left to right, pairwise, creating binary sandhi domains. In case of an odd number of syllables, the leftover syllable is absorbed into the final sandhi domain.

(7) 五 五 五
wǔ wǔ wǔ
five five five
'5-5-5'

(8) 五 五 五 五 五
wǔ wǔ wǔ wǔ wǔ
five five five five five
'5-5-5-5-5'

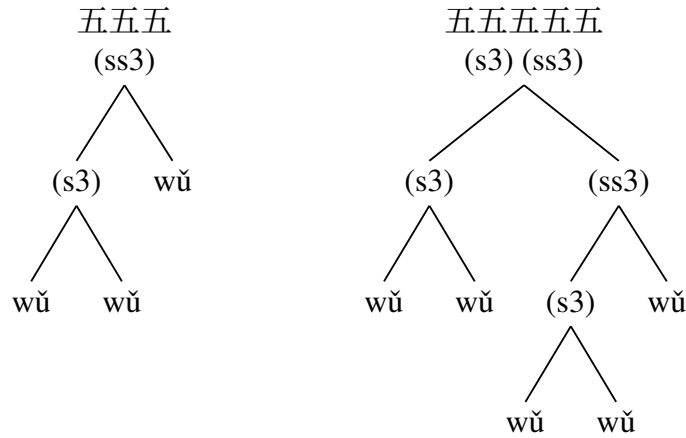


Figure 4: Typical analysis of 3TS in 'unstructured' strings, examples (7) and (8), respectively

In M. Chen's (2000) analysis of 3TS, sandhi domains are determined by minimal rhythmic units (MRUs), which are constructed according to basic metrical principles that can be overridden by the influence of morphosyntactic boundaries. These principles and said interfering influences are all encoded in constraints in the Optimality Theory (OT) framework first proposed by Prince and Smolensky (1993). An OT grammar is a ranked set of constraints such that, for any underlying input, various candidates are evaluated based on which constraints they violate, the ranking of the constraints, and the number of violations committed in order to determine the optimal surface output. In Chen's account, where no morphosyntactic structure is present, MRUs are determined by only the following three ranked constraints.

(9) BINARYITY: An MRU contains at least two syllables.

BOUNDEDNESS: An MRU contains at most two syllables.

LtoR: MRUs are constructed from left to right.

The constraints are ranked as in (10), which yields the results in the tableau in Table 2.

(10) BINARYITY $\gg$ BOUNDEDNESS $\gg$ LtoR

		BINARITY	BOUNDEDNESS	LtoR
/wu wu/ 5-5	(3)(3)	*!*		
	☞ (s 3)			
/wu wu wu/ 5-5-5	(3)(3)(3)	*! * *		
	(s 3)(3)	*!		
	(3)(s 3)	*!		*
	☞ (s s 3)		*	
/wu wu wu wu/ 5-5-5-5	(3)(s s 3)	*!		
	☞ (s 3)(s 3)			
	(s s 3)(3)	*!		*
	(s s s 3)		*!*	
/wu wu wu wu wu/ 5-5-5-5-5	(3)(s 3)(s 3)	*!		
	(s 3)(s 3)(3)	*!		*
	☞ (s 3)(s s 3)		*	
	(s s 3)(s 3)		*	*!
	(s s s s 3)		**!*	

Table 2: Chen's (2000, p. 368) OT analysis of 3TS in unstructured strings; '*' indicates a violation, '*!' indicates a fatal violation, and ☞ indicates an optimal candidate

Thus, the claim is that, where no syntactic structure interferes with the default formation of sandhi domains, the optimal sandhi patterns are as in (11).

- (11) 3 syllables: /3 3 3/ → (ss3)
4 syllables: /3 3 3 3/ → (s3)(s3)
5 syllables: /3 3 3 3 3/ → (s3)(ss3)

This view has been widely accepted for many years and is commonly repeated in contemporary literature on the subject (see, e.g., F. Chen (2020, p. 2)).

2.3 Empirical issues and research questions

The empirical claims in (11) are challenged by various data in at least two ways. Moreover, the Chen's OT account described in 2.2.2 rests on a questionable theoretical assumption.

2.3.1 The 555 fallacy

On the empirical side, while pronunciation of the digit string in (12) almost always follows the predictions in (11), the one in (13) usually does not and (14) very rarely does.

- (12) 九九五 (13) 九五九 (14) 九五五
jiǔ jiǔ wǔ jiǔ wǔ jiǔ jiǔ wǔ wǔ
'9 - 9 - 5' '9 - 5 - 9' '9 - 5 - 5'

In longer strings, similar discrepancies can be found.

- (15) 五九五五九 (16) 五五五九五
wǔ jiǔ wǔ wǔ jiǔ wǔ wǔ wǔ jiǔ wǔ
'5 - 9 - 5 - 5 - 9' '5 - 5 - 5 - 9 - 5'

In an experiment whose results are detailed in the next section, (15) was usually pronounced according to the predictions in (11), but (16) never was.

The issue at hand seems to be that an (ss3) output *appears* to be the optimal default pattern for trisyllabic strings because the string typically considered is ‘555’ (*wǔ wǔ wǔ*). However, as we shall see, while strings of the form AAX (i.e., AAA or AAB) typically follow a left-branching sandhi pattern like the one shown in Figure (3a), the right-branching pattern is heavily preferred by strings with other forms, and neither one can be considered the default pattern.

2.3.2 The 1959 problem (aka the mirror image asymmetry problem)

Strings like ‘1959’ (*yī jiǔ wǔ jiǔ*) involve four syllables, but only three third tones. The initial syllable is a first tone. This raises the question of whether the output pattern for that string will be 1(ss3) or 13(s3). A similar question can be asked about the mirror image of that string, where the non-third tone appears after the three third tones: ‘9591’ (*jiǔ wǔ jiǔ yī*).

If a sandhi domain were simply computed from left to right any time two third tones were adjacent, then we would get two applications of the rule and a three-syllable sandhi domain for each string.

(17) Naive derivations of ‘1959’ and ‘9591’

/1 3 3 3/ → 1 (s 3) 3 → 1 (s s 3)

/3 3 3 1/ → (s 3) 3 1 → (s s 3) 1

On the other hand, if (following M. Chen) a four-syllable string is divided into binary metrical units first, and sandhi is constrained by the metrical boundaries, then we would predict a two-syllable domain for each string.

(18) MRU-based derivations of 1959 and 9591

/1 3 3 3/ → (1 3) (3 3) → 1 3 (s 3)

/3 3 3 1/ → (3 3) (3 1) → (s 3) 3 1

Whichever method we choose, the strings ‘1959’ and ‘9591’ will each be predicted to have a sandhi domain of the same length. This prediction is not accurate. In reality, strings of the form /X333/ are usually realized as X3(s3), while strings of the form /333X/ are usually realized as (ss3)X.

2.3.3 Chen’s harmony problem

One remarkable thing about Chen’s treatment of basic 3TS patterns shown in Table 2 is that there is nothing in the analysis that explicitly bans adjacent third tones. For as long as third tone sandhi has been a topic of discussion, it has been considered a dissimilatory process triggered by the presence of adjacent third tones. Chen’s constraints refer only to the length of MRUs and the direction in which they are formed. The analysis therefore attempts to derive the dissimilatory effect of 3TS without directly encoding the phonological generalization that motivates the process in the first place. Deriving such a generalization from basic restrictions on the size of metrical units would be elegant if

it were empirically accurate, but the account makes incorrect predictions. Consider the typology of possible rankings of the three constraints in (9), shown in Table 3.

Constraint ranking	Optimal output
BINARITY $\gg$ BOUNDEDNESS $\gg$ LtoR	(s s 3)
BINARITY $\gg$ LtoR $\gg$ BOUNDEDNESS	(s s 3)
LtoR $\gg$ BINARITY $\gg$ BOUNDEDNESS	(s s 3)
LtoR $\gg$ BOUNDEDNESS $\gg$ BINARITY	(s 3) 3
BOUNDEDNESS $\gg$ BINARITY $\gg$ LtoR	(s 3) 3
BOUNDEDNESS $\gg$ LtoR $\gg$ BINARITY	(s 3) 3

Table 3: Optimal outputs under the six possible rankings of Chen’s (2000) constraints

The output 3(s3) is a ‘harmonically bounded’ candidate, i.e., not a possible winner under any permutation of the constraints. However, that pattern is in fact the hugely preferred one for expressions like (14), and is slightly preferred for the one in (13), which can be pronounced like a right-branching 3(s3) or a left-branching (ss3), but never as (s3)3.

The section that follows elaborates on the data that supports the empirical challenges we have briefly outlined above. The section following that proposes an analysis that avoids the ‘555 fallacy’ and the ‘1959 problem.’

3 Experiments and Findings

The data discussed here were collected during a series of three elicitation experiments designed to analyze 3TS patterns in a variety of syntactic patterns and ‘unstructured’ strings. All participants were native Mandarin speakers from various parts of Taiwan. The primary methodology utilized a storyboard-based elicitation. In Experiments 1 and 2, participants were shown a video with English narration and Mandarin subtitles. After viewing, selected scenes were shown again without audio, and participants were prompted to read the subtitles aloud. Recordings were then stored and analyzed. Experiment 3 was a similar series of elicitation tasks but required no storyboard.

3.1 Experiment 1: 3TS patterns in various structures

Our suspicion that the (ss3) pattern is not the default surface form for unstructured trisyllabic strings emerged in the course of the initial psycholinguistic experiment. While most of the prompts contained linguistic expressions with specific phrase structure, a few of them contained unstructured strings, specifically phone numbers (Figure 5). By far, the most common pronunciation of the string ‘955 995 959’ was the one shown in the spectrogram in Figure (6), where ‘955’ and ‘959’ carry the 3(s3) pattern and only ‘995’ surfaces as (ss3). Table 4 shows the data for all participants.

As can be seen in the table, the claims of C. Shih, M. Chen, F. Chen, and others that (ss3) is the default/optimal structure for trisyllabic unstructured strings of third tones was supported only by the data for the string ‘995’ and not for the other two. Based on this observation, we hypothesized that the appearance of the perceived optimality of the (ss3)



Figure 5: Examples of prompts from the original storyboard experiment; Left: ‘The phone number is 886 955 995 959’; Center: ‘Try this number: (0836) 55595’; Right: ‘Please try this number: (0836) 59559’

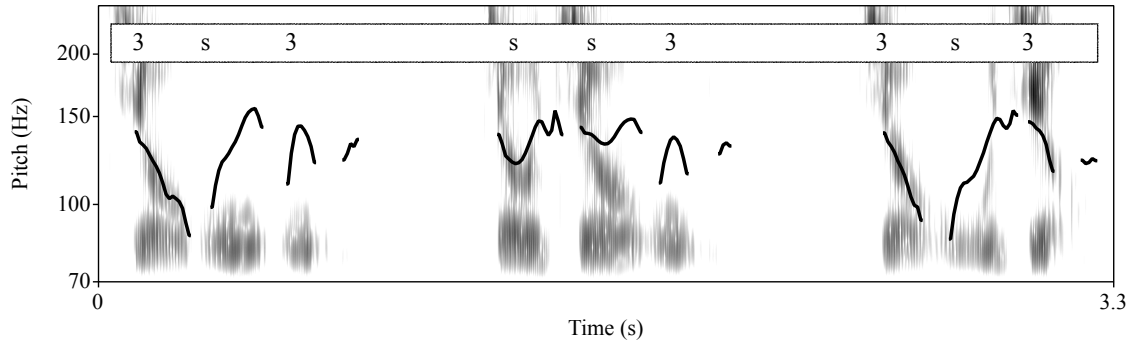


Figure 6: Spectrogram of a male participant pronouncing the phone number ‘955 995 959’

structure is somewhat of an illusion induced by the fact that the first two syllables of strings like *wǔ-wǔ-wǔ* or *wǔ-wǔ-jiǔ* are identical and that the (ss3) pattern might be optimal for strings of the form AAX (i.e., AAA or AAB) but not for trisyllabic strings of third tones in general.

We wished to further test that hypothesis. Because Mandarin has only two digits that bear the third tone (5 and 9), we would be unable to test strings of the form ABC (three distinct lexical items) if we only used numbers in the elicitation prompts. We therefore designed another experiment.

Sandhi	‘955’	‘995’	‘959’
Pattern	<i>jiǔ-wǔ-wǔ</i>	<i>jiǔ-jiǔ-wǔ</i>	<i>jiǔ-wǔ-jiǔ</i>
3(s3)	88.6	2.9	60.0
(ss3)	11.4	97.1	40.0

Table 4: Response patterns for trisyllabic strings of third-tone digits in the first elicitation experiment (in %, $n=35$)

3.2 Experiment 2: lists of non-digit syllables and X333 numerical strings

In the second experiment, users viewed a 90-second storyboard with English narration and Chinese subtitles. The narrator introduced her family and their Chinese zodiac signs: horse (馬 *mǎ*), dog (狗 *gǒu*), rat (鼠 *shǔ*), and tiger (虎 *hǔ*). These were chosen because they consist of third-tone syllables, providing the necessary lexical variety to test AAA, AAB, ABB, and ABC patterns (Table 5). They also provide the additional opportunity to test the sandhi behavior in the presence of word boundaries, since two of the zodiac signs have both monosyllabic and disyllabic variants. (For lack of space, we will not discuss the data that involved word boundaries in this paper.)

Additionally, we tested whether the patterns for trisyllabic strings of third-tone digits would maintain the sandhi patterns we observed in the first experiment if they were preceded by a non-third-tone digit, e.g, in a four-syllable expressions like ‘1959,’ ‘1999,’ etc.

<i>storyboard character</i>	<i>zodiac sign</i>	<i>word</i>	<i>pronunciation</i>
younger sister (narrator)	tiger	(老) 虎	(lǎo) hǔ
mother	rat	(老) 鼠	(lǎo) shǔ
grandmother / older sister	dog	狗	gǒu
father / grandfather / uncle	horse	馬	mǎ

Table 5: Zodiac signs and character relationships in the second storyboard experiment

Pattern Type	AAA	AAB		ABB		ABC
String	馬馬馬	馬馬狗	狗狗鼠	馬狗狗	狗馬馬	虎狗馬
333	25.0	16.7	8.3	4.2	12.5	8.3
3(s3)	4.2	0	0	87.5	70.8	45.8
(s3)3	0	0	0	0	0	0
(ss3)	70.8	83.3	91.7	8.3	16.7	45.8

Table 6: Response patterns for trisyllabic zodiac-sign strings (in %, $n=24$)

As shown in Table 6 and summarized in Table 7, there is a stark categorical shift in sandhi preference based on whether the string follows an AAX (repeated initial) or ABB (repeated final) pattern. This confirmed the findings of the first experiment that the preference for sandhi application is highly sensitive to the lexical composition of the string; the (ss3) pattern was the majority output only when strings were of the form AAX. For ABB and ABA forms, 3(s3) was preferred. Notably, the ABC pattern—consisting of three distinct lexical items—resulted in a perfect split between the two sandhi strategies.

With regard to the question of whether trisyllabic strings of third-tone digits would maintain their sandhi patterns when preceded by an additional (non-third-tone) syllable, the answer was categorically negative. See Table 8.



Figure 7: Examples of prompts from the second storyboard experiment involving trisyllabic strings of third tones preceded by a non-third-tone syllable and unstructured lists of Chinese zodiac signs

	AAA	AAB	ABB	ABB
3(s3)	5.6	0	86.1	50
(ss3)	94.4	100	13.9	50

Table 7: Summary of preference for 3(s3) vs. (ss3) across string types (in %)

3.3 Experiment 3: more four-syllable strings

Finally, a third experiment was conducted to analyze tone patterns in four-digit sequences without the use of storyboards. This phase aimed firstly to further examine unstructured strings by testing more patterns of four-digit strings (e.g., 5555, 5559, 9555, etc.) to observe sandhi application in a wider variety of sequences and, secondly, to re-test numerical patterns of the form X333 and compare the results to ‘mirror-image’ strings of the form 333X. The latter was meant (a) to compare sandhi patterns for strings with a non-third-tone digit following a string of third tones to examples with a non-third-tone digit preceding the string of third tones and (b) to re-test the findings in Experiment 2 related to X333 strings in case the pronunciation of strings denoting dates like 1955 or 1999 were special cases whose pronunciation was atypical, perhaps due to their frequency of use.

The results of the experiment eliciting utterances of four-syllable strings of third-tone-digits are given in Table 9. It is clear that the presence or absence of adjacent, identical elements has a massive influence on the sandhi pattern chosen.

Sandhi Pattern	‘995’	‘959’	‘955’	‘1995’	‘1959’	‘1955’
3(s3)	3	60	89	—	—	—
(ss3)	97	40	11	—	—	—
13(s3)	—	—	—	100	100	100
1(ss3)	—	—	—	0	0	0

Table 8: Comparison of response patterns for 3-digit and 4-digit numerical strings (in %)

Sandhi Pattern	AAAA	AABA	ABAA	ABAB	AABB	AAAB	ABBA	ABBB
(s3)(s3)	94.4	94.4	94.4	100	94.4	55.6	66.7	33.3
3(ss3)	0	0	0	0	0	0	16.7	50.0
(sss3)	5.6	5.6	5.6	0	5.6	44.4	16.7	16.7

Table 9: Response patterns for 4-syllable sequences of third-tone digits (in %, $n=18$)

To confirm the hypothesis that strings of the form X333 (where X is a non-third-tone) and 333X typically yield sandhi domains of different sizes, we used one elicitation prompt in which all the third tones were identical elements: 7555 (*qī wǔ wǔ wǔ*) versus 5557 (*wǔ wǔ wǔ qī*) to see if the identity of three adjacent syllables with the first-tone digit 7 would induce different results than what we had already observed. Conclusions from previous observations were largely confirmed, though: where the string of third-tone-syllables came first, it always formed a single trisyllabic sandhi domain, whereas when the non-third-tone syllable appeared before the string of third tones, the output was X3(s3) for all but four participants. In a prompt where the first two digits of the third-tone syllable substring did not match—7959 (*qī wǔ jiǔ wǔ jiǔ*) versus 9597 (*jiǔ wǔ jiǔ qī*)—the data were even more categorical: every elicited utterance of ‘7959’ contained a two-syllable sandhi domain, and all participants pronounced ‘9597’ with a three-syllable sandhi domain (Table 10).

Sandhi Pattern	X333	333X
X3(s3)	88.9	0
X(ss3)	11.1	0
(s3)3X	0	0
(ss3)X	0	100

Table 10: Response patterns for four-syllable numerical strings with tri-syllabic third-tone substrings (in %, $n=36$)

3.4 Conclusions

The combined data from these experiments lead to two primary conclusions regarding unstructured strings.

First, the surface pattern of a trisyllabic sequence of third tones is sensitive to its internal composition. The frequently cited (ss3) pattern is not a general default for such strings. Rather, it is strongly favored when the first two syllables are identical, as in strings of the form AAA or AAB. When only the final two syllables are identical (ABB strings), speakers instead strongly prefer the right-branching pattern 3(s3). ABC strings permit both patterns in approximately equal proportions. The apparent default status of (ss3), therefore, results largely from the traditional reliance on examples such as ‘555’, in which the first two syllables are necessarily identical. This is the empirical basis of what we have called the ‘555 fallacy.’

The results for four-syllable strings of third tones reinforced this conclusion. For while such strings show a strong overall preference for division into two binary domains, (s3)(s3), strings such as AAAB, ABBA, and ABBA permit substantially more variation, with the latter even surfacing as 3(ss3) half of the time, compared to only one third of the time as (s3)(s3).

Secondly, an unmistakable discrepancy is observed in three-syllable strings of third tones adjacent to a non-third-tone: when the non-third-tone syllable precedes three third tones, the prevailing output is X3(s3), with only a two-syllable sandhi domain. When the same sequence of third tones precedes the non-third-tone syllable, the output is invariably (ss3)X, with a three-syllable domain. Thus, /X333/ and /333X/ do not behave symmetrically, despite being mirror images of one another and containing identical substrings of adjacent third tones. This is the empirical basis of the ‘1959 problem.’

An adequate analysis of 3TS in strings that lack bona fide syntactic boundaries must account for both generalizations. This is the aim of the section that follows.

4 A Stochastic OT analysis of 3TS based on Match Theory

4.1 Introduction and goals

Our present goal is to give an analysis of 3TS that provides empirical coverage for ‘unstructured’ strings. Below we propose a structural description of these strings, followed by an analysis that avoids the ‘555 fallacy’ and the ‘1959 problem,’ discussed above. We use a probabilistic OT model—Stochastic Optimality Theory—to reflect asymmetric variation among the various patterns in the collected data.

We have already concluded that the strings *wǔ-wǔ-wǔ* (‘555’); *wǔ-wǔ-jiǔ* (‘559’); *wǔ-jiǔ-jiǔ* (‘599’); and *hǔ, gǒu, mǎ* (‘tiger, dog, horse’) do not actually have the same structure and that what distinguishes them is the presence or absence—and the position of—adjacent, identical elements that appear to be prosodically grouped together. Therefore, we will assume that adjacent identical elements are prosodically grouped into a constituent, which we designate as χ .

Our first aim will be to state a grammar that returns outputs for the various inputs of three- and four-syllable strings of third tones with approximately the same frequency as

string	<i>hǔ, gǒu, mǎ</i>	<i>wǔ-wǔ-wǔ</i>	<i>wǔ-wǔ-jiǔ</i>	<i>wǔ-jiǔ-jiǔ</i>
underlying structure	ABC	[AAA] _χ	[AA] _χ B	A[BB] _χ

Table 11: Structures of various trisyllabic strings

those in the data from the three experiments discussed in section 3 and summarized in Table 12.

Trisyllabic Pattern	3(s3)	(ss3)	
ABC	50	50	
[AAA]	1	99	
[AA]B	1	99	
A[BB]	87	13	
Quadrisyllabic Pattern	(s3)(s3)	3(ss3)	(sss3)
ABAB	100	0	0
[AAA]B	56	0	44
A[BBB]	33	50	17
A[BB]A	66	17	17

Table 12: Summary of collected data for trisyllabic and quadrisyllabic strings of third tones from Experiments 1–3 (in %)

4.2 Classical OT and a first step

Optimality Theory (OT) (Prince and Smolensky 1993; McCarthy and Prince 1993a, 1993b, 1994, 1995) was proposed as an alternative to the rule-based phonological analyses introduced by Chomsky and Halle (1968), and quickly became the dominant grammar model in generative phonology. In an OT grammar (as we have already seen in M. Chen’s analysis in section 2.2.2), a surface output, rather than being derived from an underlying representation through ordered rules, is chosen from a candidate set as the winner of a competition that compares the relative ‘harmony’ of outputs given a certain input and a set of ranked, conflicting constraints.

Constraints in OT phonology typically come in three varieties: faithfulness constraints, which require the output to preserve properties of the input; markedness constraints, which require outputs to avoid structures that are ‘marked’ or dispreferred; and alignment constraints, which require linguistic constituents or features to align with particular edges. Our analysis will begin with the following four constraints. The first three constraints are markedness constraints and the fourth is a faithfulness constraint.

(19) *33: Two 3rd tone syllables may not appear adjacent to one another.

*CC: Avoid complex contours. (I.e., avoid full 3rd tones.)²

²This constraint follows the assumption that the underlying form of Mandarin’s third tone is in fact a complex contour. Yip (1980, 2002) argues against this view and analyzes it as a low tone. If one sided with

BIN(φ): A sandhi domain must be maximally bisyllabic.

MAX-T: Every tone in the input must have a correspondent in the output.

All four of these constraints have been used in previous work by other authors.³ The complete grammar would yield a typology of $4!=24$ possible grammars, with three possible winning candidates.

/ABC/	*33	*CC	BIN(φ)	MAXT
333	*!*	* * *		
3(s3)		*!		*
(s3)3	*!	*		*
☞ (ss3)			*	**

Table 13: OT tableau for trisyllabic third-tone input using constraints in (19), with one of the twenty-four possible rankings; ‘*’ indicates a violation, ‘*!’ indicates a fatal violation, and ☞ indicates an optimal candidate

Note that the output (s3)3 is harmonically bounded, since it commits all the same violations committed by 3(s3), plus an additional one. This approach thus already avoids M. Chen’s ‘harmony problem,’ discussed in section 2.3.3.

4.3 Stochastic OT

In the Stochastic OT (StOT) of Boersma (1997), each constraint c has a ranking value $r(c) \in \mathbb{R}$. At every evaluation, independent Gaussian noise ε_c , drawn from a normal distribution $\mathcal{N}$ with a mean of zero and a standard deviation σ , is added to each $r(c)$ to yield a ‘perturbed’ value $r'(c)$.

$$r'(c) = r(c) + \varepsilon_c, \quad \varepsilon_c \sim \mathcal{N}(0, \sigma^2).$$

This makes the evaluation stochastic, where the probability of c_1 outranking c_2 is a monotonic function of their ranking difference, $r(c_1) - r(c_2)$. Because the difference between two independent normal distributions is also normally distributed, with a variance equal to the sum of their variances, and because the evaluation noise σ is typically fixed

Yip in this matter, then a constraint *Low, militating against the third tone, would work just as well for our present purposes. The unification of our analysis in this article with a more general account of the basic form and the surface alternants of the third tone must remain for now an area of further research, though see both Lee-Schoenfeld and Kandybowicz (2009) and F. Chen (2020) for related discussions.

³The constraint *33 has obvious motivations and has been used, e.g., by Zhang (1997). It is often generalized to and/or compared with the Obligatory Contour Principle (OCP) advocated by Leben (1973) and Goldsmith (1976) as a restriction on underlying representations and later by McCarthy (1986) and Yip (1988) as a principle that could mediate or trigger phonological alternation. Our BIN(φ) is identical to N. Zhang’s (1997, p. 308) constraint *Maximal Domain* as well as the constraint BINARITY of M. Chen (2000), discussed in section 2.2, though Chen’s constraint applies generally to minimal rhythmic units and not sandhi domains per se. The constraint *CC is the same constraint proposed by Lee-Schoenfeld and Kandybowicz (2009). Something like MAX-T is invoked in almost every OT analysis of tone phonology.

at 2.0 (meaning the combined variance is $2\sigma^2 = 8$), the probability of $c_1 \gg c_2$ can be calculated as:

$$P(c_1 \gg c_2) = P(\varepsilon_{c_2} - \varepsilon_{c_1} < r(c_1) - r(c_2)) = \Phi\left(\frac{r(c_1) - r(c_2)}{2\sqrt{2}}\right)$$

where Φ is the cumulative distribution function of the standard normal distribution. The probability of $c_1 \gg c_2$ is roughly depicted in Figure 8.

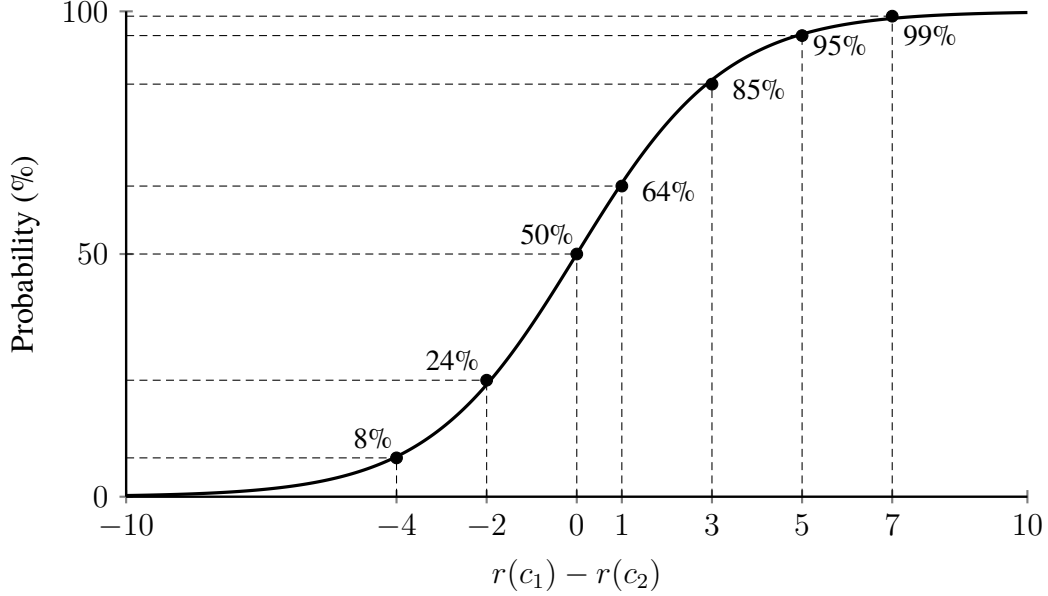


Figure 8: Probability that c_1 outranks c_2 as a function of the ranking-value difference $r(c_1) - r(c_2)$ in StOT.

While calculating exact probabilities analytically is straightforward for simple pairwise constraint conflicts, determining the precise output frequencies for a complete grammar with multiple interacting constraints quickly becomes mathematically intractable. To bypass this, researchers (e.g., Boersma and Hayes 2001) typically rely on Monte Carlo simulations (Metropolis and Ulam, 1949) to estimate predicted output probabilities. Computationally sampling the random noise for every constraint thousands of times and tallying the winning candidate on each trial yields a highly reliable approximation of the grammar’s actual output distribution. These innovations permit OT grammars to describe gradient well-formedness and asymmetric variation in ways that classical OT is not capable of.

To illustrate, Table 14 shows the results of a Monte Carlo simulation with 10,000 trials for a StOT grammar with the constraints in (19) given ranking values of 20.0; 2.0; 1.5; and 0.0, respectively.⁴

Based on the experimental findings discussed in section 3, the even split between the candidates 3(s3) and (ss3) is desirable and avoids the ‘555 fallacy’ discussed above. On the other hand, the predictions for four-syllable strings, shown in Table 15, are currently

⁴Note that we use numbers to tally violations in StOT tableaux rather than asterisks since, in a stochastic grammar, there is no way to pinpoint a ‘fatal’ violation and the asterisk notation thus has no advantage.

/ABC/	20.0 *33	2.0 *CC	1.5 Bin(φ)	0.00 MaxT	win%
333	2	3	0	0	0.00%
3(s3)	0	2	0	1	50.09%
(s3)3	1	2	0	1	0.00%
(ss3)	0	1	1	2	49.91%

Table 14: StOT tableau for trisyllabic strings with constraint ranking values and win probabilities

inaccurate since ABAB strings in our data always had an (s3)(s3) output. Moreover, the analysis currently makes no reference to χ 's and thus makes no distinction between strings like ABC, [AA]B, A[BB], etc.

/ABAB/	20.0 *33	2.0 *CC	1.5 Bin(φ)	0.0 MaxT	win%
3333	3	4	0	0	0.00%
33(s3)	1	3	0	1	0.00%
(s3)33	2	3	0	1	0.00%
3(s3)3	1	3	0	1	0.00%
(s3)(s3)	0	2	0	2	50.15%
3(ss3)	0	2	1	2	0.00%
(ss3)3	1	2	1	2	0.00%
(sss3)	0	1	2	3	49.85%

Table 15: StOT tableau with the same ranking values as in Table 14 showing incorrect predictions for a four-syllable string /ABAB/

4.4 Extending the analysis with Match Theory

We will expand our analysis to better reflect the frequency patterns in the collected data by appealing to the interaction between phonological sandhi domains (φ) and metrical feet (Ft). Unlike M. Chen, who defines sandhi domains as MRUs which are basically metrical, but can be influenced by morphosyntactic boundaries, we treat φ 's and Ft as separate domains subject to Match theory.

We will assume for this discussion that all metrical feet in Mandarin are exactly two syllables. This assumption has many precedents (though they are not always categorical). Earlier work of M. Chen (1979) proposed binary foot structure in his analysis of Chinese verse. The ‘Immediate Constituency’ clause of C. Shih’s (1986, p. 110) foot formation rules, proposed to account for Mandarin prosody, explicitly demands the formation of binary units. Duanmu (1993, 2007, 2022) has analyzed stress in Mandarin by assuming trochaic (i.e., disyllabic or bimoraic) structure and argued that disyllables in Mandarin invariably form prosodic feet. Feng (2000) proposes that the ‘natural foot’ in modern spoken Chinese is disyllabic, and that trisyllabic units are ‘super feet.’ Very recently, Zhuang et al. (2025) have given a historical description of the “foot-shift” that took place

in the evolution from Old Chinese’s monosyllabic feet to the disyllabic feet of Modern Mandarin.

The striking discrepancies among trisyllabic strings of third tones appearing in three- and four-syllable strings (cf. 995 versus 1995) shown in Table 8 is strong evidence that sandhi domains are sensitive to metrical structure: AAB, ABA, and ABB exhibit variation but XAAB, XABA, and XABB strings do not. We claim that the presence or absence of alternation is partly due to the presence or absence of multiple foot structures. I.e., two distinct footing patterns are available for a trisyllabic string, namely $\{xx\}x$ and $x\{xx\}$, whereas a four-syllable string must be footed as $\{xx\}\{xx\}$.

Match Theory (Selkirk, 2011) uses two types of alignment constraints called Match and Wrap to relate phonological structure with syntax. We will use it to regulate the interaction between metrical feet (Ft), sandhi domains (φ) and substrings (χ) that form internal structural components of numbers and lists.

(20) $WRAP(\chi, \varphi)$: Every χ must be contained in a single sandhi domain φ .

$MATCH(\chi, \varphi)$: For every χ , there must be a sandhi domain φ that consists of the exact same set of syllables.

$WRAP(Ft, \varphi)$: Every metrical foot must be contained in a single sandhi domain φ .

$MATCH(Ft, \varphi)$: For every metrical foot, there must be a sandhi domain φ that consists of the exact same set of syllables.

The Match and Wrap constraints in (20) allow us to capture the strong preference for (ss3) outputs for [AA]B strings where χ is on the left, and the preference for 3(s3) when it is on the opposite side.⁵ When the Wrap constraints were given a ranking value of 10 and the Match constraints a value of 5, the simulation shown in Table 16 returned a 3(s3) sandhi pattern with a matching foot structure, $x\{xx\}$, for an A[BB] input 94% of the time. Such a foot structure and sandhi pattern, given that input, satisfies all Wrap and Match constraints. On the other hand, for an [AA]B input, the only output that satisfies all the Wrap and Match constraints is $x\{xx\} / (s3)3$, which violates the highly ranked *33 constraint. Thus, the output frequencies for an [AA]B input were about evenly split

⁵It is necessary to mention F. Chen (2020) here, who uses Match Theory in her analysis of the syntax/phonology interface surrounding 3TS, with particular attention to the asymmetry of left- versus right-branching structures. As her principal concern is the alignment of tone-sandhi domains (φ) with syntactic phrases (XP), her Match-theoretic approach is closely related to the analysis developed here. We lack the space to give a full comparison of our analysis to hers, but the most important theoretical difference is that Chen permits recursive sandhi domains. Thus, a phrase such as [mǎi [hǎo jiǔ]] ‘buy good wine’ may receive the structure $(mai_3 (hao_s jiu_3)_\varphi)_\varphi$, in which the VP and the embedded object each corresponds to a distinct φ (see *Ibid.*, p. 6). The postulation of a φ corresponding to each recursively embedded XP allows constraints like $Match(\varphi, XP)$ and $Match(XP, \varphi)$ to be satisfied rather easily. We see no reason to accept that a trisyllabic output of the form 3(s3) has more than one sandhi domain, and based on the definition of this concept that we gave in the introduction, the present analysis assumes that any syllable can belong to at most one φ , whose boundaries will be identifiable in the obvious way. Empirically, the analysis Chen advocates predicts that the sandhi pattern (sss3) is the only possible output for a double left-branching input like [zǎo diǎn [zǒu [hǎo]]] (‘Leaving a bit early is good’). On her account, the output (s3)(s3) for that sentence is not only not optimal, but harmonically bounded, since (sss3) would not violate any constraints (*Ibid.* example 21). Our experimental data did not confirm this prediction; in an elicitation experiment with that exact sentence, the results were an even split between both sandhi patterns. I leave further comparison as a matter of future research.

between the two possible foot structures and almost always (ss3). This is a fairly accurate reflection of the data for strings of these forms observed in our experiments.

/A[BB]/	20.0 *33	10.0 WRAP(χ, φ)	10.0 WRAP(Ft, φ)	5.0 MATCH(χ, φ)	5.0 MATCH(Ft, φ)	2.0 *CC	1.5 BIN(φ)	0.0 MAXT	win%
{xx}x 333	2	1	1	1	1	3	0	0	0.0%
{xx}x 3(s3)	0	0	1	0	1	2	0	1	0.0%
{xx}x (s3)3	1	1	0	1	0	2	0	1	0.0%
{xx}x (ss3)	0	0	0	1	1	1	1	2	3.0%
x{xx} 333	2	1	1	1	1	3	0	0	0.0%
x{xx} 3(s3)	0	0	0	0	0	2	0	1	94.0%
x{xx} (s3)3	1	1	1	1	1	2	0	1	0.0%
x{xx} (ss3)	0	0	0	1	1	1	1	2	3.0%

/[AA]B/	20.0 *33	10.0 WRAP(χ, φ)	10.0 WRAP(Ft, φ)	5.0 MATCH(χ, φ)	5.0 MATCH(Ft, φ)	2.0 *CC	1.5 BIN(φ)	0.0 MAXT	win%
{xx}x 333	2	1	1	1	1	3	0	0	0.0%
{xx}x 3(s3)	0	1	1	1	1	2	0	1	0.0%
{xx}x (s3)3	1	0	0	0	0	2	0	1	0.0%
{xx}x (ss3)	0	0	0	1	1	1	1	2	48.3%
x{xx} 333	2	1	1	1	1	3	0	0	0.0%
x{xx} 3(s3)	0	1	0	1	0	2	0	1	3.4%
x{xx} (s3)3	1	0	1	0	1	2	0	1	0.0%
x{xx} (ss3)	0	0	0	1	1	1	1	2	48.3%

Table 16: StOT tableau for [AA]B and A[BB] with each foot structure specified in output candidates and Match/Wrap constraints. The optimal output for A[BB] was 3(s3) 94% of the time. The optimal output for [AA]B was (ss3) 96.6% of the time.

Unfortunately, the predictions for other strings are not yet as accurate. E.g., the same grammar predicts that an ABC input returns a 3(s3) pattern almost 90% of the time, spoiling the accurate 50-50 split that we had originally achieved. Moreover, for strings of the form [AAA]B, A[BBB], and A[BB]A, the (sss3) pattern would be the overwhelming favorite since this output always satisfies both Wrap constraints. To address the latter issue, we propose a constraint that further limits the extent of sandhi domains. Three-syllable sandhi domains in Mandarin are very common. Four-syllable domains are far less common, and domains larger than 5 syllables are very rare.

- (21) *BRANCH(φ , Ft): A φ may not branch into multiple metrical feet.
(One violation for each complete metrical foot it contains beyond the first.)

To restore predictions of an approximately equal frequency of two- versus three-syllable sandhi domains for strings of the form ABC, we propose two amendments. First, we suppose that Mandarin prefers to foot expressions from the left edge rather than leaving the first syllable extrametrical. This is similar in spirit to the ‘Duple Meter’ clause of C. Shih’s (1986, p. 110) foot formation rules that demand left-to-right formation and to M. Chen’s LtoR constraint, which penalizes the failure to begin forming metrical groups from the beginning of an expression, discussed in section 2.2.

- (22) ALIGN-L(ι , Ft): The left edge of an intonational phrase must be aligned with the left edge of a metrical foot.

The constraint ALIGN-L(ι , Ft) will favor {xx}x footing over x{xx}. Where an ABC string is footed as {xx}x, it will almost always surface as (ss3) due to the influence of

$\text{WRAP}(\text{Ft}, \varphi)$. If we suppose the constraint to have a ranking value equal to those of the two Match constraints, this would result in a more balanced distribution of the two sandhi patterns for an ABC string. On the other hand, it would also disrupt our predictions for strings with the form A[BB], which would be predicted to surface as (ss3) about one-third of the time, when, in the experimental data, it was less than half that often. The latter prediction could be improved by increasing the ranking of $\text{MATCH}(\chi, \varphi)$ from 5 to 8. Full tableaux for both trisyllabic and quadrisyllabic strings with the final constraint ranking are shown in the Appendix.

4.5 A summary of simulation outputs vs. experimental results

A comparison of the predictions of the StOT grammar we have proposed and the frequencies manifested in the data we collected for three- and four-syllable strings of third-tone syllables in the three elicitation experiments discussed in section 3 are shown in Table 17.

3-SYLLABLE STRINGS						
Input	Predictions		Experimental Data			
	3(s3)	(ss3)	3(s3)	(ss3)		
ABC	49%	51%	50%	50%		
[AAA]	2%	98%	1%	99%		
[AA]B	3%	97%	1%	99%		
A[BB]	89%	11%	87%	13%		

4-SYLLABLE STRINGS						
Input	Predictions			Experimental Data		
	(s3)(s3)	3(ss3)	(sss3)	(s3)(s3)	3(ss3)	(sss3)
ABAB	100%	0%	0%	100%	0%	0%
[AAA]B	49%	0%	51%	56%	0%	44%
A[BBB]	32%	43%	24%	33%	50%	17%
A[BB]A	34%	33%	33%	66%	17%	17%

Table 17: Comparison of outputs predicted by the proposed grammar and the data for three- and four-syllable strings collected from elicitation experiments

The predictions fall within the 95% Wilson confidence intervals (Wilson, 1927) for the experimental proportions in all but one category. The sole exception is the A[BB]A string. For this input, our grammar predicts an approximately even distribution across three candidate outputs, whereas the experimental data showed a preference for (s3)(s3) over the other two, selected by approximately two-thirds of the participants. One possible interpretation is that the presence of identical adjacent elements exerts less influence on tone-sandhi phrasing when those elements occur medially than when they occur at the edge of an expression. The nature of this apparent positional discrepancy remains an issue for further research.

4.6 The 1959 problem: quadrisyllabic expressions with trisyllabic third-tone substrings

In addition to the results above, the grammar proposed here predicts the positional asymmetry that gives rise to the ‘1959/mirror image asymmetry problem’ described in section 2.3.2. Recall that strings of the form X333 and 333X do not receive parallel sandhi parses: X333 strings overwhelmingly favored a final binary sandhi domain, X3(s3), whereas 333X strings categorically favored a trisyllabic domain, (ss3)X. The simple left-to-right application of 3TS over pairs of third-tone syllables made opposite predictions to that of left-to-right grouping over pairs of syllables in general, followed by 3TS, and both of those strategies were only half right. The present account makes correct predictions for both cases:

For an input of the form /X333/, both X3(s3) and X(ss3) sandhi patterns will violate $\text{WRAP}(\text{Ft}, \varphi)$ one time. The candidate X3(s3) performs better compared to its rival in terms of the constraint $\text{MATCH}(\text{Ft}, \varphi)$, whereas X(ss3) is the better performer in terms of the constraint *CC. But *CC is ranked significantly lower than $\text{MATCH}(\text{Ft}, \varphi)$, so this circumstance favors X3(s3). Moreover, X3(s3) also outperforms X(ss3) in terms of the constraint MAXT . For these reasons, the X3(s3) output wins almost 88% of the time for the input /X333/, but X(ss3) is still optimal about 12% of the time.

For input of the form /333X/, the story is different. The sandhi domain in the output 3(s3)X doesn’t fully contain either of the metrical feet in the expression and thus violates $\text{WRAP}(\text{Ft}, \varphi)$ twice. The output (ss3)X only violates that constraint once. The other competitor that violates $\text{WRAP}(\text{Ft}, \varphi)$ only once is (s3)3X. But that candidate offends the highly ranked *33 constraint. For the above reasons, (ss3)X wins almost 100% of the time.

/X333/	20.0 *33	10.0 *Br(φ , Ft)	10.0 W(χ , φ)	10.0 W(Ft, φ)	8.0 M(χ , φ)	5.0 M(Ft, φ)	5.0 A-L(ι , Ft)	2.0 *CC	1.5 Bin(φ)	0.0 MaxT	win%
{xx}{xx} X333	2	0	0	2	0	2	0	3	0	0	0.0%
{xx}{xx} X3(s3)	0	0	0	1	0	1	0	2	0	1	87.9%
{xx}{xx} X(s3)3	1	0	0	2	0	2	0	2	0	1	0.0%
{xx}{xx} X(ss3)	0	0	0	1	0	2	0	1	1	2	12.1%

/333X/	*33	*Br(φ , Ft)	W(χ , φ)	W(Ft, φ)	M(χ , φ)	M(Ft, φ)	A-L(ι , Ft)	*CC	Bin(φ)	MaxT	win%
{xx}{xx} 333X	2	0	0	2	0	2	0	3	0	0	0.0%
{xx}{xx} 3(s3)X	0	0	0	2	0	2	0	2	0	1	0.2%
{xx}{xx} (s3)3X	1	0	0	1	0	1	0	2	0	1	0.0%
{xx}{xx} (ss3)X	0	0	0	1	0	2	0	1	1	2	99.8%

Table 18: Comparison outputs for strings of the form X333 versus 333X, where X is a non-third tone

Statistically, the frequencies output by the simulation match the experimental data very nicely; they are all within a single percentage point of the response patterns reported in section 3, Table 10.

5 Conclusion

The data presented in this article demonstrated that Mandarin third tone sandhi (3TS) in strings of digits and lists—traditionally categorized as ‘unstructured’—is in fact heavily influenced by the nature and position of elements in the strings, especially whether there are substrings of repeating syllables in the string and where in the string those substrings appear. We have shown that 3TS patterns can be more accurately modeled when these substrings of repeating elements (such as those in ‘559’ or ‘9995’) are treated as prosodic constituents.

We showed how a grammar with two pairs of constraints from Match Theory could regulate the correspondence of these constituents with two other prosodic categories: sandhi domains themselves and binary metrical feet. In conjunction with a few familiar markedness and faithfulness constraints that set the foundation for the behavior of third tones and their tendency to dissimilate and form sandhi domains, as well as an additional alignment constraint favoring left-aligned foot structures, the grammar was able to reflect the statistical patterns in the experimental data with remarkable accuracy when combined with the appropriate ranking values and the flexibility afforded by Stochastic Optimality Theory.

The same grammar was also able to account for the asymmetry exhibited by sandhi patterns that occur in strings with an extra non-third-tone syllable, such as in the discrepancy among typical domains in strings like ‘1959’ (X333) versus the mirror image of that string ‘9591’ (333X).

While the present analysis offers a new foundation for analyzing strings lacking morphosyntactic structure, many areas for further inquiry remain. We are optimistic that the foundation suggested above can be extended and elaborated on in ways that will improve our understanding of Mandarin third tone sandhi and aspects of the syntax/phonology interface that are involved in it.

Appendix: Simulation Results

Table 19: Simulation results for 3-syllable strings ABC, [AAA], [AA]B, and A[BB]

		20.0 *33	10.0 *Br(φ ,Ft)	10.0 W(Ft, φ)	10.0 W(χ , φ)	8.0 M(χ , φ)	5.0 M(Ft, φ)	5.0 Al-L(Ft, ι)	2.0 *CC	1.5 Bin φ	0.0 MaxT	win%
/ABC/	333	2	0	1	0	0	1	0	3	0	0	0.0%
{xx}x	3(s3)	0	0	1	0	0	1	0	2	0	1	0.1%
	(s3)3	1	0	0	0	0	0	0	2	0	1	0.0%
	(ss3)	0	0	0	0	0	1	0	1	1	2	51.1%
x{xx}	333	2	0	1	0	0	1	1	3	0	0	0.0%
	3(s3)	0	0	0	0	0	0	1	2	0	1	48.8%
	(s3)3	1	0	1	0	0	1	1	2	0	1	0.0%
	(ss3)	0	0	0	0	0	1	1	1	1	2	0.0%
/[AAA]/	333	2	0	1	1	1	1	0	3	0	0	0.0%
{xx}x	3(s3)	0	0	1	1	1	1	0	2	0	1	0.0%
	(s3)3	1	0	0	1	1	0	0	2	0	1	0.0%
	(ss3)	0	0	0	0	0	1	0	1	1	2	98.3%
x{xx}	333	2	0	1	1	1	1	1	3	0	0	0.0%
	3(s3)	0	0	0	1	1	0	1	2	0	1	1.7%
	(s3)3	1	0	1	1	1	1	1	2	0	1	0.0%
	(ss3)	0	0	0	0	0	1	1	1	1	2	0.0%
/[AA]B/	333	2	0	1	1	1	1	0	3	0	0	0.0%
{xx}x	3(s3)	0	0	1	1	1	1	0	2	0	1	0.0%
	(s3)3	1	0	0	0	0	0	0	2	0	1	0.0%
	(ss3)	0	0	0	0	1	1	0	1	1	2	97.0%
x{xx}	333	2	0	1	1	1	1	1	3	0	0	0.0%
	3(s3)	0	0	0	1	1	0	1	2	0	1	3.0%
	(s3)3	1	0	1	0	0	1	1	2	0	1	0.0%
	(ss3)	0	0	0	0	1	1	1	1	1	2	0.0%
/A[BB]/	333	2	0	1	1	1	1	0	3	0	0	0.0%
{xx}x	3(s3)	0	0	1	0	0	1	0	2	0	1	2.0%
	(s3)3	1	0	0	1	1	0	0	2	0	1	0.0%
	(ss3)	0	0	0	0	1	1	0	1	1	2	10.7%
x{xx}	333	2	0	1	1	1	1	1	3	0	0	0.0%
	3(s3)	0	0	0	0	0	0	1	2	0	1	87.3%
	(s3)3	1	0	1	1	1	1	1	2	0	1	0.0%
	(ss3)	0	0	0	0	1	1	1	1	1	2	0.0%

Table 20: Simulation results for 4-syllable strings ABAB, [AAA]B, A[BBB], and A[BB]A

		20.0	10.0	10.0	10.0	8.0	5.0	5.0	2.0	1.5	0.0	
		*33	*Br(φ ,Ft)	W(Ft, φ)	W(χ , φ)	M(χ , φ)	M(Ft, φ)	Al-L(Ft, ι)	*CC	Bin φ	MaxT	win%
/ABAB/ {xx}{xx}	3333	3	0	2	0	0	2	0	4	0	0	0.0%
	33(s3)	1	0	1	0	0	1	0	3	0	1	0.0%
	(s3)33	2	0	1	0	0	1	0	3	0	1	0.0%
	3(s3)3	1	0	2	0	0	2	0	3	0	1	0.0%
	(s3)(s3)	0	0	0	0	0	0	0	2	0	2	99.7%
	3(ss3)	0	0	1	0	0	2	0	2	1	2	0.0%
	(ss3)3	1	0	1	0	0	2	0	2	1	2	0.0%
	(sss3)	0	1	0	0	0	2	0	1	2	3	0.3%
/[AAA]B/ {xx}{xx}	3333	3	0	2	1	1	2	0	4	0	0	0.0%
	33(s3)	1	0	1	1	1	1	0	3	0	1	0.0%
	(s3)33	2	0	1	1	1	1	0	3	0	1	0.0%
	3(s3)3	1	0	2	1	1	2	0	3	0	1	0.0%
	(s3)(s3)	0	0	0	1	1	0	0	2	0	2	49.1%
	3(ss3)	0	0	1	1	1	2	0	2	1	2	0.0%
	(ss3)3	1	0	1	0	0	2	0	2	1	2	0.0%
	(sss3)	0	1	0	0	1	2	0	1	2	3	50.9%
/A[BBB]/ {xx}{xx}	3333	3	0	2	1	1	2	0	4	0	0	0.0%
	33(s3)	1	0	1	1	1	1	0	3	0	1	0.0%
	(s3)33	2	0	1	1	1	1	0	3	0	1	0.0%
	3(s3)3	1	0	2	1	1	2	0	3	0	1	0.0%
	(s3)(s3)	0	0	0	1	1	0	0	2	0	2	32.4%
	3(ss3)	0	0	1	0	0	2	0	2	1	2	43.2%
	(ss3)3	1	0	1	1	1	2	0	2	1	2	0.0%
	(sss3)	0	1	0	0	1	2	0	1	2	3	24.4%
/A[BB]A/ {xx}{xx}	3333	3	0	2	1	1	2	0	4	0	0	0.0%
	33(s3)	1	0	1	1	1	1	0	3	0	1	0.0%
	(s3)33	2	0	1	1	1	1	0	3	0	1	0.0%
	3(s3)3	1	0	2	0	0	2	0	3	0	1	0.0%
	(s3)(s3)	0	0	0	1	1	0	0	2	0	2	34.1%
	3(ss3)	0	0	1	0	1	2	0	2	1	2	32.7%
	(ss3)3	1	0	1	0	1	2	0	2	1	2	0.0%
	(sss3)	0	1	0	0	1	2	0	1	2	3	33.2%

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